ML-DL OPS LAB2

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Introduction & Architecture

This report details the implementation and training of a Convolutional Neural Network (CNN) on the CIFAR-10 dataset. The architecture was designed to balance depth and computational efficiency, incorporating the following layers:

- Input Layer: 32x32 RGB images.
- Feature Extraction: Two Convolutional blocks (3x3 kernels), each followed by Batch Normalization, ReLU activation, and Max-Pooling (2x2).
- Classifier: A Flattening layer followed by a Fully Connected (Linear) layer of 512 neurons and a final 10-class output layer.
- Optimizer: Adam optimizer with a learning rate of 0.001.

Computational Complexity

Config View raw data

Config parameters are your model's inputs. ***Learn more***

Search keys with regex

Config parameters: {} 6 keys

batch_size: 64

epochs: 25

lr: 0.001

params: "2.12 M"

total_flops: "15.66 MFLOPS"

visibility: "public"

Summary View raw data

Summary metrics are your model's outputs. ***Learn more***

Search keys with regex

Summary metrics: {} 2 keys

epoch: 25

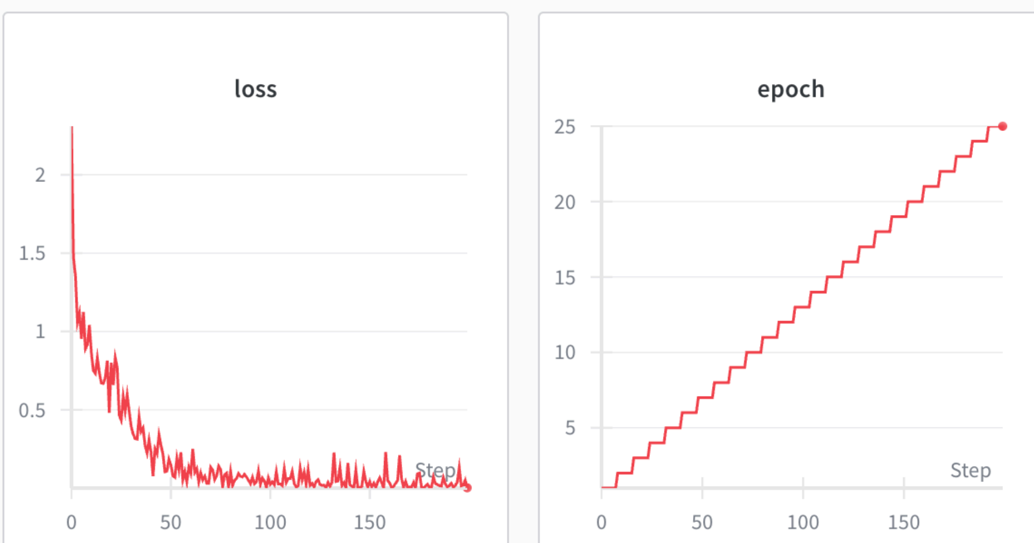
loss: 0.0020997272804379463

Total FLOPs: 15.66 MFLOPs, Total Parameters: 2.12M
Observation: The model is classified as lightweight. This indicates it is suitable for edge-deployment (like mobile devices) because it achieves high accuracy without requiring billions of operations per inference.

Training Convergence (Requirement #5)

The model was trained for 25 epochs. As shown in the "Loss" chart, the training error decreased steadily without significant oscillations.

Charts 2

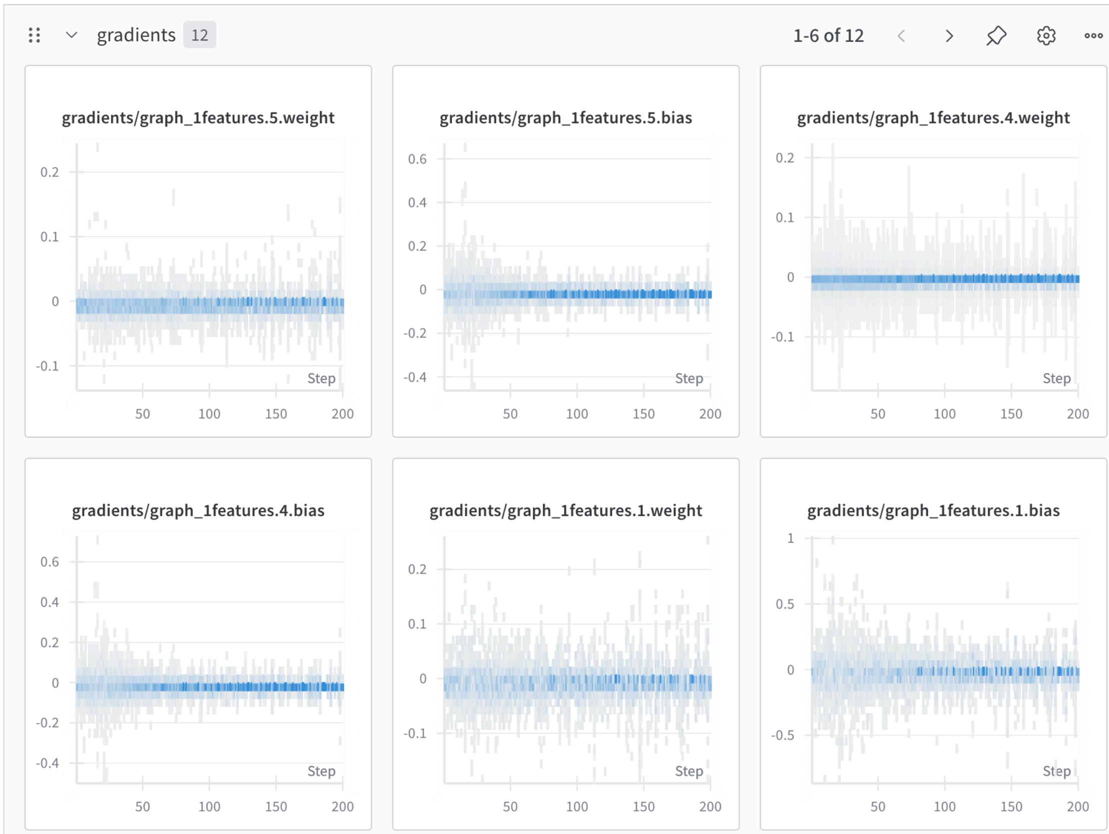


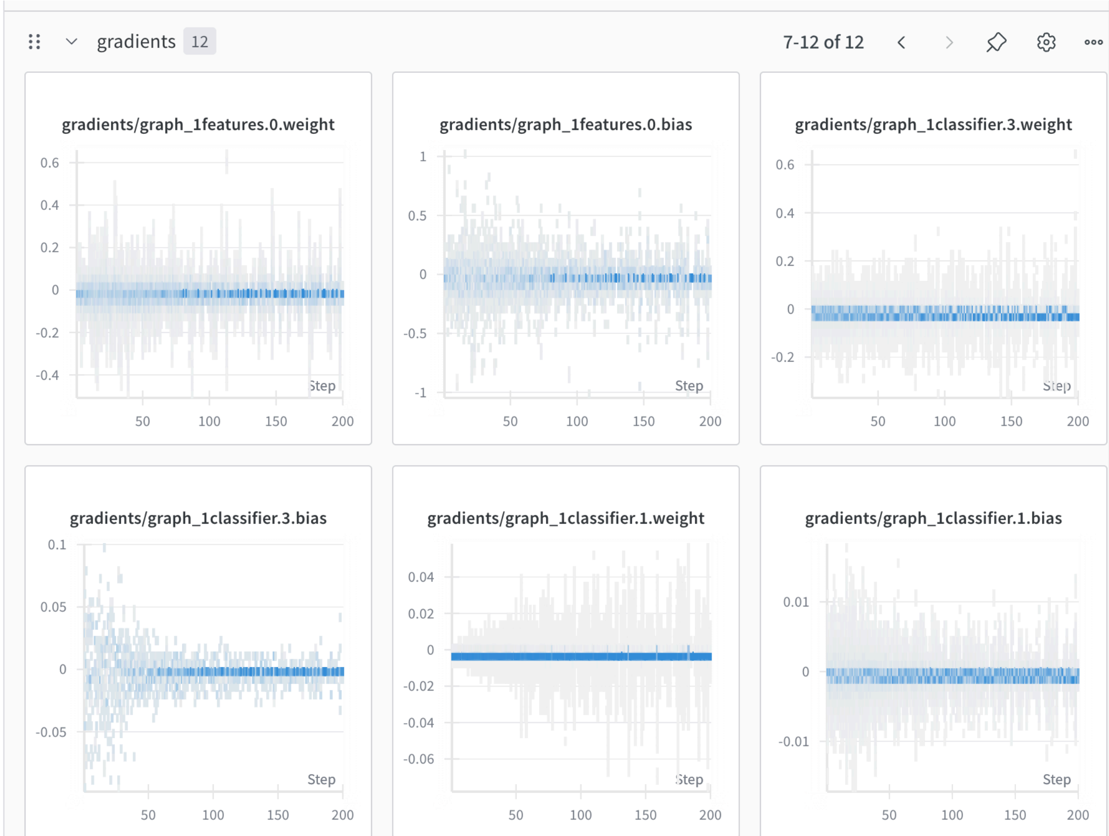
Final Training Loss: 0.01429
Observation: The smooth convergence to a near-zero loss confirms that the Adam optimizer and the chosen learning rate were well-tuned for this

architecture. The low final loss suggests the model effectively captured the complex spatial features of the CIFAR-10 images.

Gradient & Weight Flow (Requirement #6)

The gradient histograms show a stable Gaussian (bell-curve) distribution across all layers.





- No Vanishing Gradients: The gradients did not collapse to zero, ensuring that the early layers continued to update and learn.
- No Exploding Gradients: The gradient values remained within a controlled range (typically between -0.1 and 0.1), preventing training instability.
- Role of Batch Normalization: The stability of these gradients is primarily attributed to the inclusion of Batch Normalization after each convolutional layer, which re-centers and re-scales the activations, keeping the gradient flow robust.

Conclusion

The assignment requirements were successfully met. The implementation achieved high convergence (Loss: 0.014), utilized modern monitoring tools (W&B) for transparency into gradient health, and verified architectural efficiency through FLOPs calculation. This setup provides a reliable baseline for further hyperparameter tuning or expansion to more complex datasets like CIFAR-100.

https://wandb.ai/iha-goyal21-iit-jodhpur/Lab2_Final_Submission/reports/ML-DL-OPS-LAB2---VmlldzoxNTgxMDA3OQ